

Formal R&D project proposal / statement of work

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Project objective

Opyjo Inc. will determine whether a privacy-preserving, content-agnostic adaptive engine can infer reliable learner state from sparse outcomes and select practice that improves durable learning across tenants.

Technical uncertainties

The project will test sparse-data mastery estimation, correction of item-difficulty priors from behaviour, durable outcome improvement over a fixed baseline, and safe generalization across different curricula and traffic patterns.

Systematic work

WP1 compares EWMA, Elo/IRT and confidence-weighted learner models. WP2 develops empirical item calibration and drift controls. WP3 conducts a stable adaptive-versus-baseline outcome experiment. WP4 validates isolation, latency, coverage and mapping with external tenants.

Milestones and deliverables

M1 model-selection report and versioned parameters (Month 3); M2 calibration pipeline and review thresholds (Month 5); M3 controlled outcome report (Month 8); M4 external-tenant validation and onboarding playbook (Month 10).

Success measures

Calibration error, log-loss, cold-start accuracy, prior-versus-observed difficulty error, delayed retention, time-to-mastery, recommendation coverage, p95 latency, tenant-isolation tests, and reproducible model decisions.